Hyperspectral Band Selection for Spectral–Spatial Anomaly Detection

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***Abstract*— Owing to significantly improved spectral resolution, a hyperspectral imaging sensor can now uncover many unknown subtle material substances. In many cases, anomalies are usually embedded in the background. To develop a means through which these anomalies may be detected and separated from the background, we propose a spectral–spatial anomaly detection method based on a selected band subset. To be specific, we constrain an unsupervised network by making full use of the underlying physical characteristics which are beneficial to hyperspectral anomaly detection. Based on that, a selection criterion is constructed to adaptively select a subset of bands that essentially contain discriminative and informative features between the anomaly and background in an unsupervised manner. Then, the selected bands are simultaneously inputted into the spatial detector and spectral detector. To overcome the deficiencies of detecting anomalies in only one aspect, an adaptive combination of spatial result and the spectral result is introduced. Finally, a simple and powerful iterative suppression is conducted on the initial detection map to further reduce false alarm rate while ensuring detection capability. Extensive empirical researches performed on eighteen publicly available hyperspectral images (HSIs) of different sizes over different scenes demonstrate that our proposed method can achieve an average detection capability of 0.99564, and the average false alarm rate is one order of magnitude lower than the second one.**

***Index Terms*— Anomaly detection, band selection, hyperspectral image (HSI), spectral–spatial optimization, unsupervised representation learning.**

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I. Introduction

HYPERSPECTRAL imaging sensors can capture subtle spectral characteristics of the materials in an image scene. Owing to the strong advantage of hyperspectral image (HSI), it can be scientifically and quantitatively applied in many domains. However, the problems of insufficient samples, interference bands, and data imbalance seriously restrict the practical application of high-dimensional HSIs. Especially for anomaly detection without any prior knowledge.

Hyperspectral anomaly detection is the detection of outliers whose spectral and spatial signatures are significantly distinct from their surrounding data samples or the global background in an unsupervised way [1]. In practice, anomalies are of particular interest in military surveillance, environmental monitoring, and agriculture [2]–[4]. For example, a tank hidden in a forest can be discovered by its spectral properties rather than spatial attributes. Because tanks and vegetation are made of different materials, it is possible to distinguish a tank hidden in a forest from vegetation. If the tank is disguised as the same color as the vegetation in the forest, it will be difficult to distinguish the tank from the forest in the spatial domain.

As a hot topic in the field of HSI applications, hyperspectral anomaly detection has received extensive attention and in-depth research. A classic detector is Reed-Xiaoli detector (RXD) [5], which calculates the Mahalanobis distance (MD) to measure the distance between a pixel and the background by assuming that the background obeys a multivariate Gaussian distribution. On the basis of RXD, numerous variants have been reported in the literature. For example, the Kernel-RX detector (KRXD) [6] extended RXD to the feature space associated with the original input space by a nonlinear mapping function, which could be regarded as a nonlinear version of RXD. The clustering-based KRXD (CKRX) [7] clustered background pixels based on KRXD, and then applied the fast feature decomposition to detect anomalous objects. Chang *et al*. [8] designed an optimized projection function to separate background and anomalies.

Besides, sparse representation theory-based methods are also applied to hyperspectral anomaly detection. The collaborative-representation-based detector (CRD) [9]

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assumed that the background could be represented by a linear combination of its spatial neighborhoods, while anomalous pixels cannot. In [10], a sparse fraction estimation framework was proposed for anomaly detection. The low-rank sparse representation (LRASR) algorithm [11] decomposed background and anomalies by adopting low-rank sparse constraints. Zhang *et al*. [12] decomposed HSIs into background, anomaly, and noise by assuming that background is of low rank and anomaly is sparse. Then, the anomalous objects are detected according to MD. A combination of spectral unmixing and low-rank matrix decomposition was also used to detect anomalies [13] whose solution is quite complicated. Whether the sparse or collaborative representation-based method, its performance is highly correlated with the dictionary used. If the background dictionary is contaminated by anomalies, the detection rate may be low.

Different from the previous methods based on spectral signatures, the spatial features such as area information and local smoothness are also exploited to achieve anomaly detection [14], [15]. In order to achieve a comprehensive result, Li *et al*. [16] linearly combined the spectral information based on the structured sparse representation model and the spatial information based on the intra-cluster reconstruction model. Besides, an autoencoder (AE) [17] is adopted to extract spatial-spectral feature matrices for anomaly detection [18], [19]. As mentioned above, all the work shows that considering spatial and spectral features together can achieve better performance on anomaly detection than just considering one. Although the combination of spatial detection and spectral detection can improve the detection accuracy, it is generally complex compared to just spatial detection or spectral detection. In particular, HSIs generally have huge data volumes and also contain vast amounts of spectral information that is highly correlated among bands. Therefore, it is crucial to select an optimal band subset representing the entire image cube by removing redundant and interference bands for spectral–spatial anomaly detection.

Since band selection can provide discriminative information and reduce the computational burden, it has received considerable interest in various applications. Unlike band selection for target detection and classification [20], [21], band selection for anomaly detection is more challenging because there is no prior knowledge that can be used to find an optimal subset of bands. Yuan *et al*. [22] presented a multigraph determinantal point process (MDPP) mode to find the best band subset in an unsupervised way. Wang *et al*. [23] proposed a band subset selection method for anomaly detection, which simultaneously selected multiple bands. Although there is no prior information available to discover anomalous features in practice, it is still necessary to further exploit the underlying characteristics in HSIs to select a subset of the bands with significant differences between the anomaly and background. Besides, how to fully combine the spatial detection and spectral detection on the selected band subset is the key to reduce the false alarm rate while maintaining detection capability.

To address the problems mentioned above, we propose a spectro-spatial anomaly detection method on the selected subset of bands that mainly contain discriminative information between the anomaly and the background. Specifically, we constrain the hidden features of an unsupervised network through its underlying characteristics in HSIs. As a consequence, the extracted features include the intrinsic properties of HSIs, which are beneficial to anomaly detection. Based on that, a band selection criterion is constructed to adaptively select a more discriminative band subset in an unsupervised manner. The intuitive analysis indicates that the selected band subset based on the physical constraint network contains more discriminative features between the anomaly and the background than the discarded bands, which have a positive influence on subsequent detection and reduce the computational burden caused by high-dimensional HSIs. To overcome the deficiency of anomaly detection in only one aspect, an adaptive method combining spatial detection and spectral detection is introduced. Besides, a simple and powerful iterative suppression is proposed to further reduce false alarm rate while retaining detection capability. Experimental results are conducted on eighteen publicly available HSIs of different sizes over different scenes, which demonstrate the superior performance of the proposed anomaly detection method in terms of detection accuracy and false alarm rate for all data sets.

The rest of this article is organized as follows. A comprehensive review on the unsupervised networks for HSI application is presented in Section II. The proposed method is specifically described in Section III. The experimental results are given in Section IV. Finally, the conclusions are drawn in Section V.

II. Background

Because of the limited samples, unsupervised learning has become an increasing interest in the analysis of high-dimensional HSIs. Traditionally used unsupervised learning algorithms mainly include principal component analysis (PCA), independent component analysis (ICA), and kernelized PCA, which have limitations on their capability to characterize rare anomalies embedded in complex background [24]–[27]. It is worth noting that the unsupervised networks such as deep belief network (DBN) [28], [29], AE [17], and variational AE (VAE) [30] can extract hidden features underlying HSIs in a low-dimensional space. Among them, AE and its variants have been widely used to detect anomalies in videos [31], [32]. AE contains an encoder and a decoder in which the parameters (weight matrix and bias) are estimated by minimizing the loss between the input data samples and the reconstruction ones

$$\min_{\mathbf{W},\mathbf{b},\mathbf{W}',\mathbf{b}'} \left[\|\mathbf{x}-\hat{\mathbf{x}}\|_2^2\right] \tag{1}$$

where $\mathbf{W}$ and $\mathbf{b}$ are the weight matrix and bias of encoder, and $\mathbf{W}'$ and $\mathbf{b}'$ are the weight matrix and bias of decoder. All the parameters are updated by the minibatch stochastic gradient descent (SGD) [33] algorithm with no requirement of prior information.

Besides the reconstruction loss, VAE particularly contains KL-Divergence regularization term, which aims at optimizing the variational lower bound on the marginal log-likelihood of each observation. Recently, Su *et al*. [34] employed VAE to perform hyperspectral unmixing, which demonstrated superior performance compared to the state-of-the-art methods. Consequently, a specifically designed VAE can help

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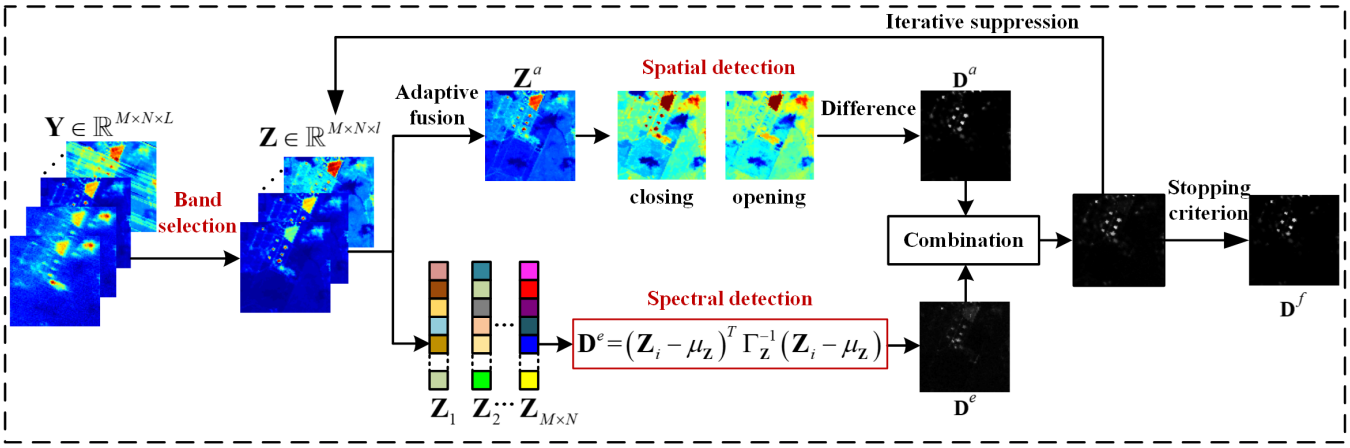


Fig. 1. Architecture for hyperspectral anomaly detection with a spectro–spatial joint optimization scheme.

to represent HSIs in deep latent space. Therefore, we apply and develop a VAE approach to exploit the underlying properties of hyperspectral data. Because of the insufficiency of physical constraints such as spectral consistency and the use of multivariate Gaussian distribution, VAE may have limited capability to analyze HSIs. Accordingly, we investigate hidden information for its suitability and practicality in hyperspectral anomaly detection.

III. Proposed Method

A schematic of the proposed anomaly detection framework is illustrated in Fig. 1. Let $\mathbf{Y} \in \mathbb{R}^{M\times N\times L}$ represent the input HSI and $\hat{\mathbf{Y}} \in \mathbb{R}^{M\times N\times L}$ represent the reconstructed one produced by the proposed decoder, where L is the total number of bands in HSI. The selected band subset is represented by $\mathbf{Z} \in \mathbb{R}^{M\times N\times l}$, where l denotes the number of the selected bands. Detailed descriptions of each step are described in Sections III-A and III-B.

A. Band Selection

Since anomalous targets are detected with no prior knowledge, an optimal subset must be selected to fully represent the original set and also appropriately preserve physical significance. So, the key to success is to construct an effective band selection rule, which is beneficial to detect anomalous targets. Interestingly, the representation-leaning networks can represent high-dimensional data in deep latent space. Huber-Lerner *et al.* [35] validated the distinct correlation between Gaussianity and anomaly detection performance. Therefore, we take advantage of Gaussianity to extract features in accordance with HSIs so as to yield better performance for anomaly detection. To be more specific, spectral angle mapper (SAM) [36], [37] is imposed on the reconstruction loss of VAE to measure the error between the reconstructed spectral vector, $\mathbf{Y}_i \in \mathbb{R}^{1\times L}$, and the input spectral vector, $\hat{\mathbf{Y}}_i \in \mathbb{R}^{1\times L}$, where the reconstruction loss can be defined by jointly combining least square with SAM value as follows:

$$\ell_r(\mathbf{Y}_i, \hat{\mathbf{Y}}_i) = \|\mathbf{Y}_i - \hat{\mathbf{Y}}_i\|_2^2 + \lambda \cos^{-1}\left(\frac{\mathbf{Y}_i \hat{\mathbf{Y}}_i^T}{\|\mathbf{Y}_i\|_2 \|\hat{\mathbf{Y}}_i\|_2}\right). \quad (2)$$

The output of the encoder, feature matrix $\mathbf{F}_i$, is sampled from a Gaussian distribution prior, i.e., $q(\mathbf{F}_i|\mathbf{Y}_i) \sim N(\mu, \sigma^2 I)$.

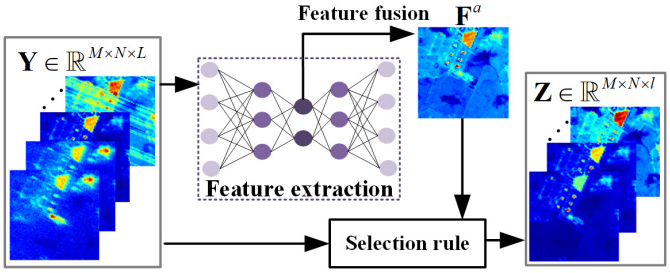


Fig. 2. Detailed description of the proposed band selection method based on the physical constraint VAE.

The parameters of encoder $(\mathbf{W}, \mathbf{b})$ and decoder $(\mathbf{W}', \mathbf{b}')$ can be estimated by minimizing the following loss:

$$\min_{\mathbf{W},\mathbf{b},\mathbf{W}',\mathbf{b}'} \|\mathbf{Y}_i - \hat{\mathbf{Y}}_i\|_2^2 + \lambda \cos^{-1}\left(\frac{\mathbf{Y}_i \hat{\mathbf{Y}}_i^T}{\|\mathbf{Y}_i\|_2 \|\hat{\mathbf{Y}}_i\|_2}\right) + KL(q(\mathbf{F}_i|\mathbf{Y}_i)\|p(\mathbf{F}_i)) \quad (3)$$

where λ is the parameter to control the influence of SAM on loss function, which is empirically set to 0.05 for all test HSIs considering the stability of network convergence. $q(\mathbf{F}_i|\mathbf{Y}_i)$ represents the posterior distribution of encoder and $p(\mathbf{F}_i)$ is the prior distribution.

The third term is the KL-Divergence regularization term promoting the distribution of $\mathbf{F}_i$ to approach Gaussian distribution. When the loss function converges, the feature matrix $\mathbf{F}$ generated by the encoder can be used to reconstruct HSI that approximates the input through the decoder. Because of the limited samples in HSIs, over-fitting may occur and need to be avoided when the network is too deep. Furthermore, extensive experiments also show that a two-layer unsupervised network can extract effective features for HSIs [38]. Therefore, we set the depth of the physical constraint VAE to 2, and the number of nodes is 20, i.e., the extracted feature matrix $\mathbf{F}$ includes 20 dimensions. Although this default network architecture may not yield the best performances for all sample images, it can provide stable and acceptable performances for most of sample images. Then, a comprehensive feature map is obtained by considering different pieces of information provided by each dimension of $\mathbf{F}$ as shown in Fig. 2. Using the strong representational capability of structure tensor developed in [39],

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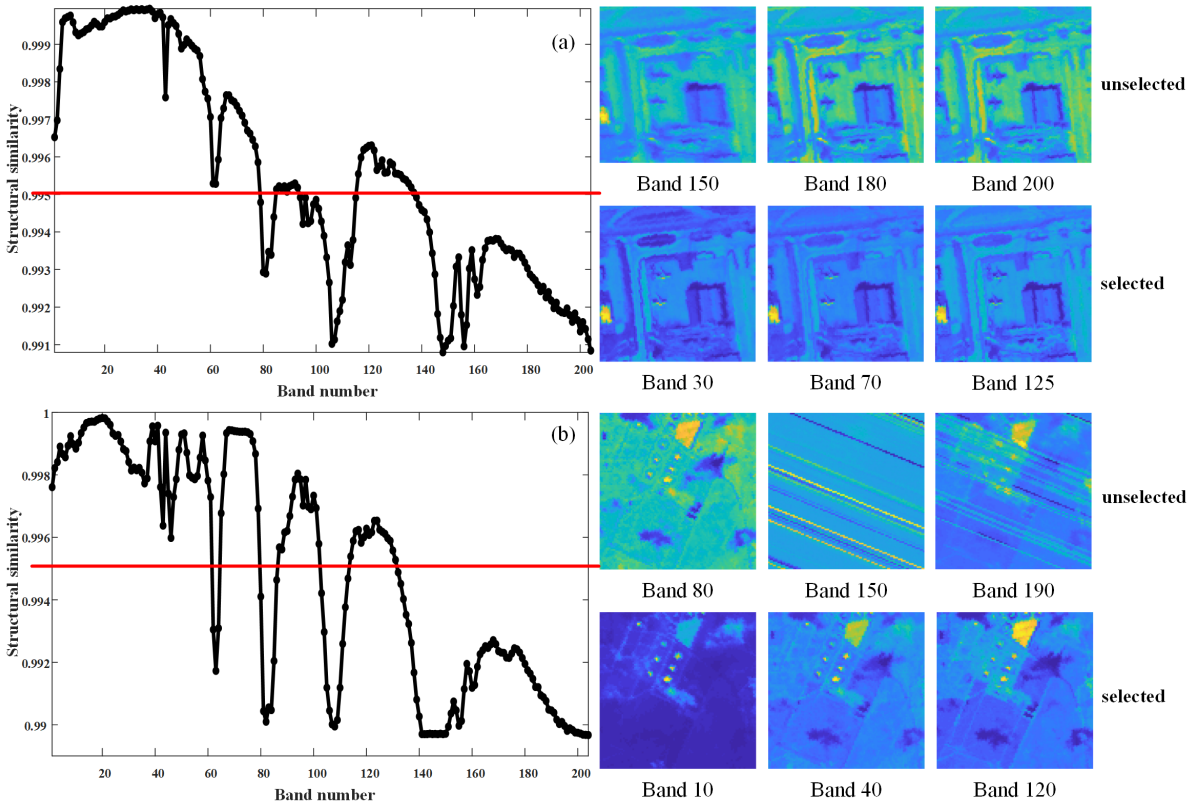


Fig. 3. Structural similarity between the fused feature map and each band of the original HSIs, and the corresponding selected and unselected bands in (a) ABU-airport2 scene and (b) ABU-urban1 scene.

we adopt an adaptive weighting method based on it

$$\mathbf{F}^a = \frac{\sum_{k=1}^{K} a_k \mathbf{F}_k}{\sum_{k=1}^{K} a_k} \tag{4}$$

where a_k is the larger eigenvalue of the structure tensor matrix from the kth node $\mathbf{F}_k$. $\sum_{k=1}^{K} a_k$ means the sum of the larger eigenvalues of the structure tensor matrix from all features. As detailed in [39], the larger eigenvalue of the structure tensor matrix can illustrate the edge intensity. Hence, we consider it as the weight coefficient of each feature node to obtain a comprehensive feature. After obtaining the fused feature map $\mathbf{F}^a$, a simple selection criterion is constructed to measure the structural similarity between each band of the original HSI and $\mathbf{F}^a$

$$\{\widehat{i}\} = \frac{(2\mu_{\mathbf{Y}^i}\mu_{\mathbf{F}^a} + c_1)(2\sigma_{\mathbf{Y}^i\mathbf{F}^a} + c_2)}{(\mu_{\mathbf{Y}^i}^2 + \mu_{\mathbf{F}^a}^2 + c_1)(\sigma_{\mathbf{Y}^i}^2 + \sigma_{\mathbf{F}^a}^2 + c_2)} \geq 0.995 \tag{5}$$

where $\mathbf{Y}^i$ represents the ith band of $\mathbf{Y}$, $\mu_{\mathbf{Y}^i}$ and $\mu_{\mathbf{F}^a}$ represent the mean of $\mathbf{Y}^i$ and $\mathbf{F}^a$, $\sigma_{\mathbf{Y}^i}^2$ and $\sigma_{\mathbf{F}^a}^2$ are the variance of these two images, $\sigma_{\mathbf{Y}^i\mathbf{F}^a}$ denotes the covariance between these two images. According to [40], the values of c_1 and c_2 are found to be $(0.01 \times g)^2$ and $(0.03 \times g)^2$, respectively, where g represents the dynamic range of pixel values. For a complete justification, we refer the reader to the work [40]. In general, the mean, variance, and covariance of an image can reflect the brightness, contrast, and structural information of an image, respectively. The greater the structural similarity, the more similar the two representations. The optimal value of structural similarity is equal to 1. If the structural similarity between $\mathbf{Y}^i$ and $\mathbf{F}^a$ is greater than or equal to a threshold, 0.995, which illustrates that the information contained in $\mathbf{Y}^i$ and $\mathbf{F}^a$ is very similar. In this case, the ith band is selected and the sequence number is set to $\widehat{i}$. All bands satisfying this condition are determined as the optimal subset $\mathbf{Z} \in \mathbb{R}^{M\times N\times l}$.

To intuitively explain the selection criterion based on structural similarity, Fig. 3 shows the structural similarity between each band $\mathbf{Y}^i$ and the fused feature map $\mathbf{F}^a$. In addition, the corresponding selected and unselected bands are also shown in Fig. 3. Obviously, when the value of structural similarity is greater, the anomaly and background have greater spectral discrimination. On the contrary, the spectra of anomaly and background are very similar. It can also be observed from Fig. 3 that the anomalies in the unselected bands have a low spectral response. This phenomenon explains that the band subset with a large gap between anomaly and background can be effectively selected based on the structural similarity between $\mathbf{Y}^i$ and $\mathbf{F}^a$.

B. Anomaly Detection

This section focuses on anomaly detection in both spatial and spectral regions. To reduce the rate of miss detection, the detection methods used in spatial and spectral domains are committed to detect anomalies with different characteristics as many as possible. Following the features of anomalies in HSIs, the spatial detection method is devoted to detecting anomalies with structural information embedded in the local homogeneous background, whereas spectral detection mainly aims at discovering anomalies with different spectra from the

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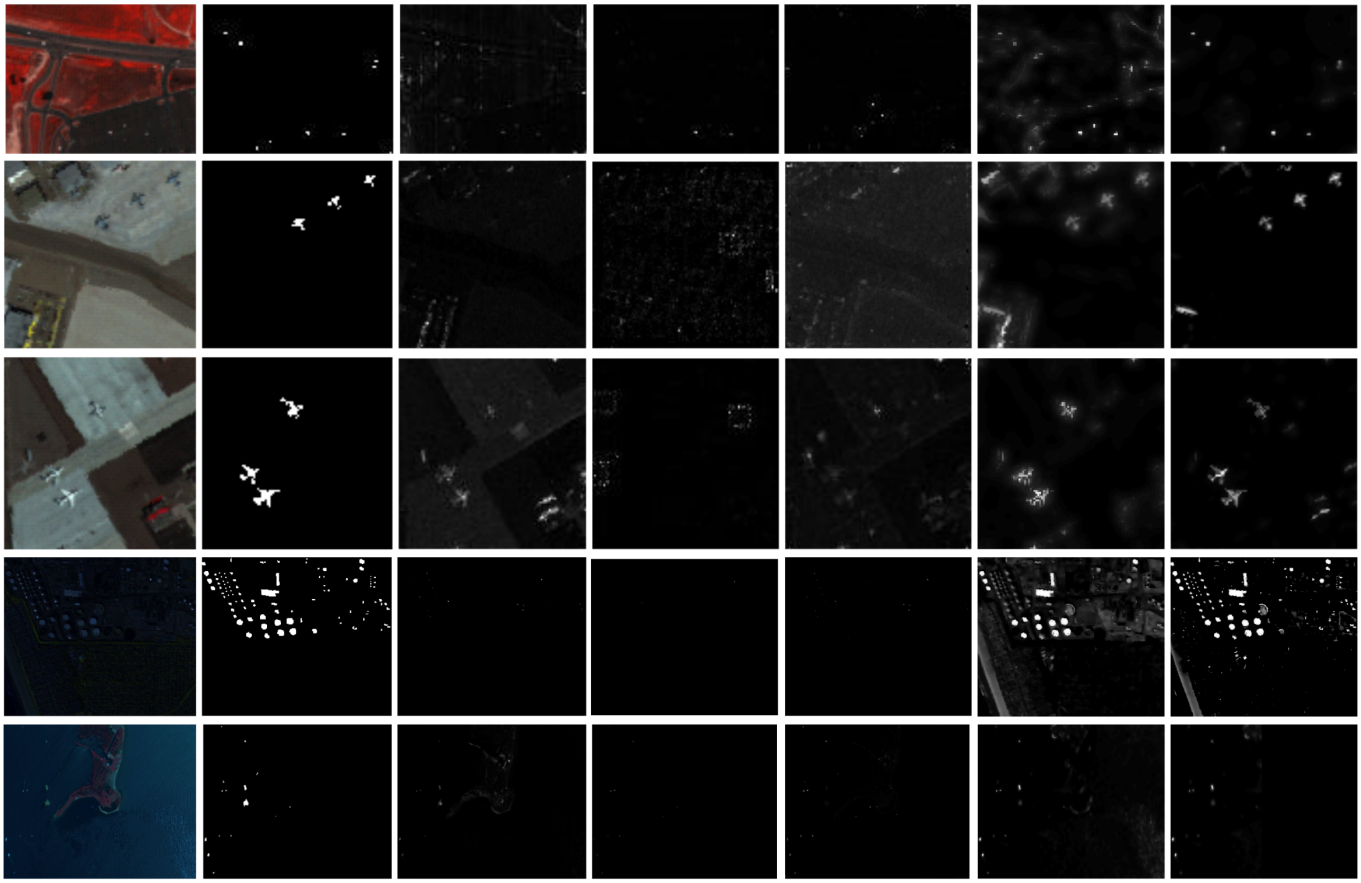


Fig. 4. Detection maps of different methods. (a) Color composites of HYDICE, two HSIs from San Diego, EI Segundo, and Grand Isle data sets, (b) reference map, (c) RX, (d) LRX, (e) CRD, (f) STGF, and (g) proposed.

background. For spatial detection, the selected band subset is first fused based on the structure tensor matrix owing to its simplicity and effectiveness. Then, the morphological closing and opening operations are utilized to extract anomalies in small areas as shown in Fig. 1. Considering the local smoothness, a guided filter is utilized to obtain the spatial detection result $\mathbf{D}^a$. More details can be found in [39]. As shown in Fig. 3, the spectral difference between anomaly and background is more prominent in the selected band subset. For spectral detection, it is simple and efficient to implement MD on the optimal subset rather than the full bands. For the spectral detection, the result can be represented as follows:

$$\mathbf{D}^e = (\mathbf{Z}_i - \mu_{\mathbf{Z}})^T \Gamma_{\mathbf{Z}}^{-1} (\mathbf{Z}_i - \mu_{\mathbf{Z}}), \quad i = 1, \ldots, MN \tag{6}$$

where $\mathbf{Z}_i \in \mathbb{R}^{l\times 1}$ is the ith vector in the selected band subset $\mathbf{Z}$, $\mu_{\mathbf{Z}} \in \mathbb{R}^{l\times 1}$ and $\Gamma_{\mathbf{Z}} \in \mathbb{R}^{l\times l}$ are the mean vector and the covariance matrix of $\mathbf{Z}$. Of particular interest are the mean vector and covariance matrix which are calculated on the band subset rather than the full bands. As a result, the negative influence of redundant information and interference bands on anomaly detection can be effectively reduced, while performance degradation due to high-dimensional features under limited samples can also be avoided. In order to take full advantage of spatial and spectral detection results, $\mathbf{D}^a$ and $\mathbf{D}^e$ are linearly combined to distinguish anomaly from background precisely by the following approach:

$$\mathbf{D}_1 = \frac{\nu_a}{\nu_a + \nu_e}\mathbf{D}^a + \frac{\nu_e}{\nu_a + \nu_e}\mathbf{D}^e \tag{7}$$

where ν_a and ν_e are the larger eigenvalues of the structure tensor matrix from $\mathbf{D}^a$ and $\mathbf{D}^e$, respectively. Most importantly, background suppression is a crucial factor to decrease the false alarm rate. Moreover, anomaly detection is universally assessed through visual effect because of no prior knowledge as reference. As stated earlier, it is a significant step for evaluating the performance of anomaly detection to suppress background. Hence, an iterative suppression method is introduced to reduce falsely alarmed targets. As illustrated in Fig. 1, the initial detection map $\mathbf{D}_1$ containing more anomaly information rather than background is multiplied by the selected band subset $\mathbf{Z}$, and the proposed detection procedure is repeated until the following stopping criterion is met:

$$\|\mathbf{D}_f - \mathbf{D}_{f-1}\|_2 \leq 10^{-4}. \tag{8}$$

The stop criterion is to calculate the distance between the previous detection result and the current result which is less than 10^{-4}.

IV. Experimental Results and Analysis

A. Data Sets Description

Several widely used data sets are used to test the proposed method (see Table I), which were captured by a diverse range

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TABLE I
DATA SETS USED FOR THE EXPERIMENTS

Dataset	Images	Pixels	Sensor
HYDICE	1	80×100	HYDICE
San Diego	2	100×100	AVIRIS
ABU	13	100×100 or 150×150	AVIRIS or ROSIS
EI Segundo	1	250×300	AVIRIS
Grand Isle	1	300×480	AVIRIS

of hyperspectral imaging sensors, including Hyperspectral Digital Imagery Collection Experiment (HYDICE), Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), and Reflective Optics System Imaging Spectrometer (ROSIS-03). Noisy water absorption bands were removed before performing experiments. An urban area captured by HYDICE includes 21 anomalous pixels with 175 bands remained. There are two images in San Diego airport area, and each image contains three aircraft with 189 bands preserved. There are three scenes including airport, beach, and urban in the ABU data set. These sample images contain 100 × 100 or 150 × 150 pixels with different band numbers such as 102, 191, 204, 205, 188, 193, and 207. EI Segundo data set has 224 spectral bands, which was captured over an area of EI Segundo, CA, USA. The image scene is composed of an area of an oil refinery, several residential areas, parks, and one school zone. The construction of an oil refinery such as storage tanks and the towers are considered as anomaly targets. The last data set was captured at the location of Grand Isle on the Gulf Coast, part of Jefferson Parish, LA, USA. Those man-made objects in the water are selected as the anomalies. The anomalies embedded in these data sets appear in the sub-pixel level as well as with structural information in different scales, which can illustrate the universality of the compared methods.

B. Detection Performance

To evaluate the performance of the proposed method, comparative experiments are conducted on four representative methods, i.e., the RXD, local RXD (LRXD), CRD, structure tensor, and guided filter (STGF)-based detector [39]. These methods have shown good performances on several real hyperspectral data sets. For the LRXD, the window size is a significant parameter. There is a difference in detection performance with different window sizes. By considering the stable performance, the sizes of two windows are 7 and 13 for other data sets except the EI Segundo data set, respectively. Such parameter settings is too inaccurate for the EI Segundo data set, so we reset the inner and the outer windows for this data set as 15 and 23, respectively. For the CRD method, the optimal regularization parameter is set to 10^{-6} according to the original literature [9]. In our experiments, the optimal parameter settings of the competitive methods for each data set are selected according to the detection performance.

The receiver operating characteristic (ROC) and the area under the curve (AUC) metric [41] are widely used objective evaluation indexes. The ROC curve, with its origin in communication theory [42], describes detection performance with true-positive rate (P_D) plotted as a function of false-positive rate (P_F) at each threshold (t). P_D defines the number of correct positive detections occur among all positive samples, and P_F defines the number of incorrect positive detections occur among all negative samples available. The ROC curve of (P_D, P_F) characterizes the tradeoff between P_D and P_F. The AUC metric can identify general trends in detection performance. The larger the AUC value of the ROC curve of (P_D, P_F), the better the performance. Specifically, the ROC curve of (P_F, t) can illustrate the degree of background suppression. The smaller the AUC value of the ROC curve of (P_F, t), the better the background suppression and the lower the false alarm rate. The ideal AUC value of the ROC curve of (P_D, P_F) is 1 and the ideal AUC value of the ROC curve of (P_F, t) is 0.

Extensive experiments are conducted on the HYDICE, San Diego, EI Segundo, Grand Isle, and ABU data sets. The detection maps obtained by the compared methods with the optimal parameters are visually demonstrated in Figs. 4–7. By observing the visual detection maps, the proposed method can better detect anomalies of different sizes captured by different sensors while suppressing background more strongly. For example, in Fig. 4, the anomalies in whether the HYDICE urban scene or the San Diego scene can be precisely detected by the proposed method and STGF method. However, the STGF method tends to emphasize background information in these examples, and thus, increasing their false alarm rates. The RXD, LRXD, and CRD methods generally do not perform detection performances for some sample images, such as Fig. 5 well. In comparison to the competitive methods, the proposed method can effectively detect anomalies of different sizes in different scenes and suppress background. For example, the structural information of three aircrafts in Fig. 5 is very clearly preserved and the background is suppressed effectively by our proposed method. By contrast, other methods cannot detect the shape of the aircraft of small size while reducing the false alarm rate.

Generally, it is difficult to reduce false alarm rate while tending to better detection performance since there is no prior knowledge for detecting anomalous objects. In order to illustrate the phenomenon objectively, we list the average AUC values of the ROC curve of (P_D, P_F) and (P_F, t) in Tables II and III, respectively. The reported average AUC value of (P_D, P_F) on all data sets supports that the proposed method tends to be the best detection performance and the STGF method ranks the second place. Besides, the average AUC value of (P_F, t) reported in Table III supports that the proposed method achieves the lowest false alarm rate on average, while the STGF method has the worst false alarm rate. Although the STGF method can also achieve competitive detection performance, the false alarm rate of this method is worse, which is consistent with the visual detection map illustrated in Figs. 4–7. It is able to conclude from Tables II and III that many methods ensure detection accuracy at the expense of background suppression and our proposed method performs better background suppression while maintaining anomaly detection capability.

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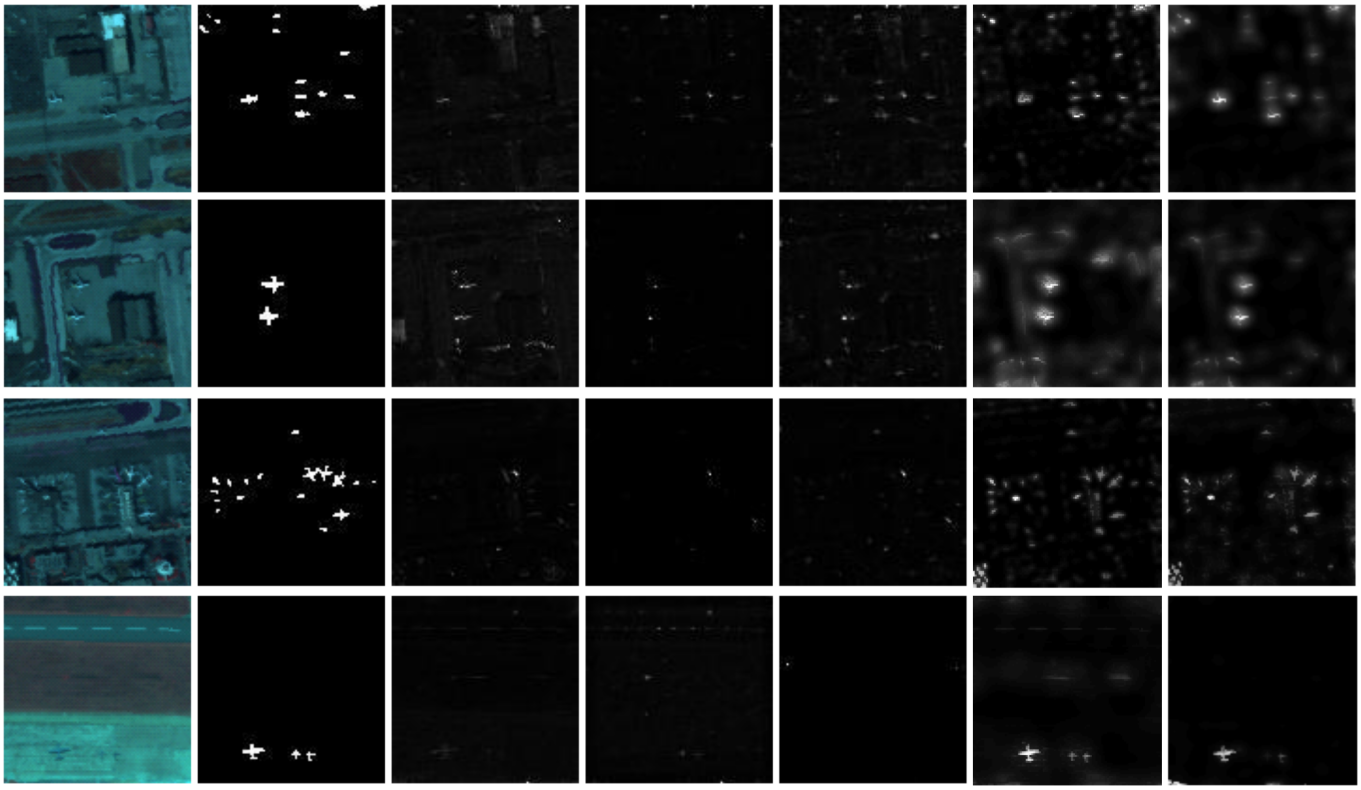


Fig. 5. Detection maps of different methods: (a) Color composites of ABU-airport scene, (b) reference map, (c) RX, (d) LRX, (e) CRD, (f) STGF, and (g) proposed.

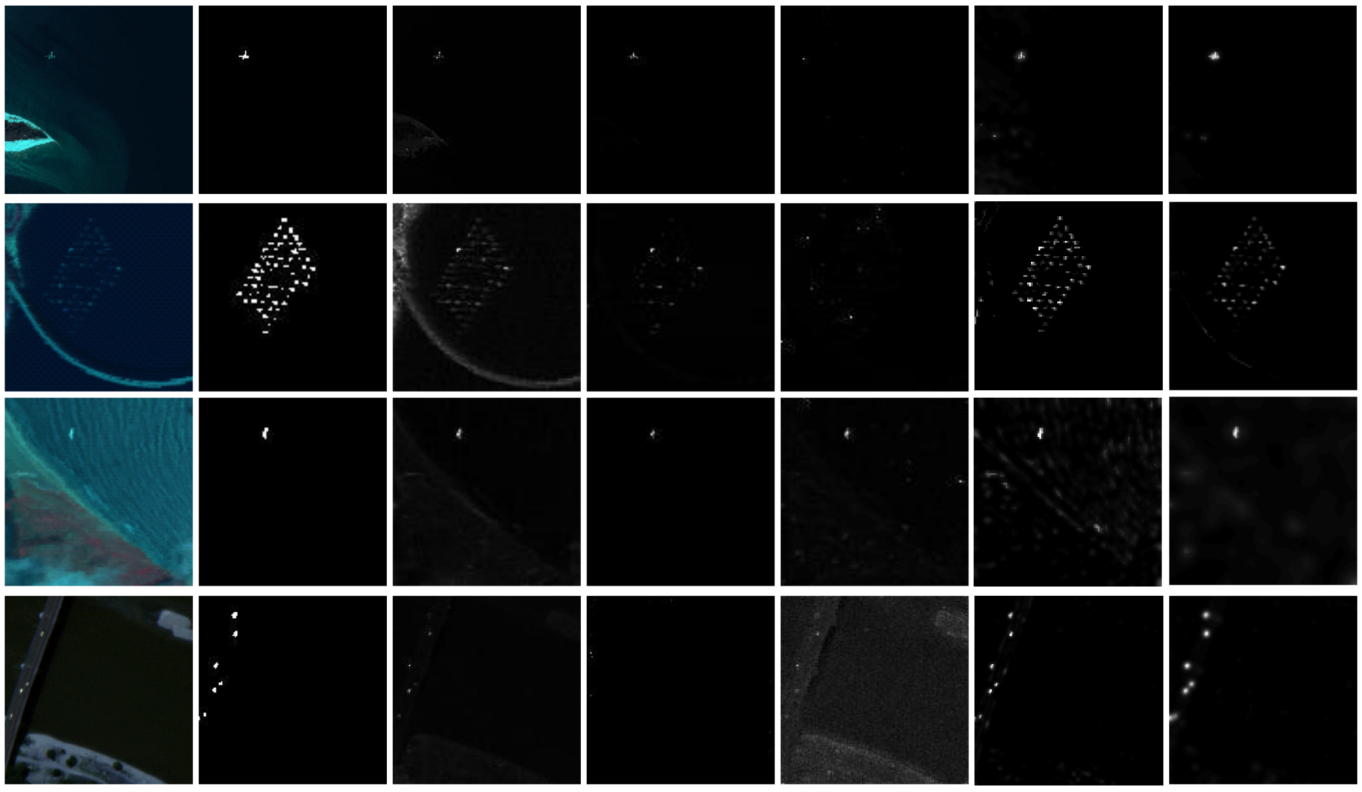


Fig. 6. Detection maps of different methods. (a) Color composites of ABU-beach scene, (b) reference map, (c) RXD, (d) LRXD, (e) CRD, (f) STGF, and (g) proposed.

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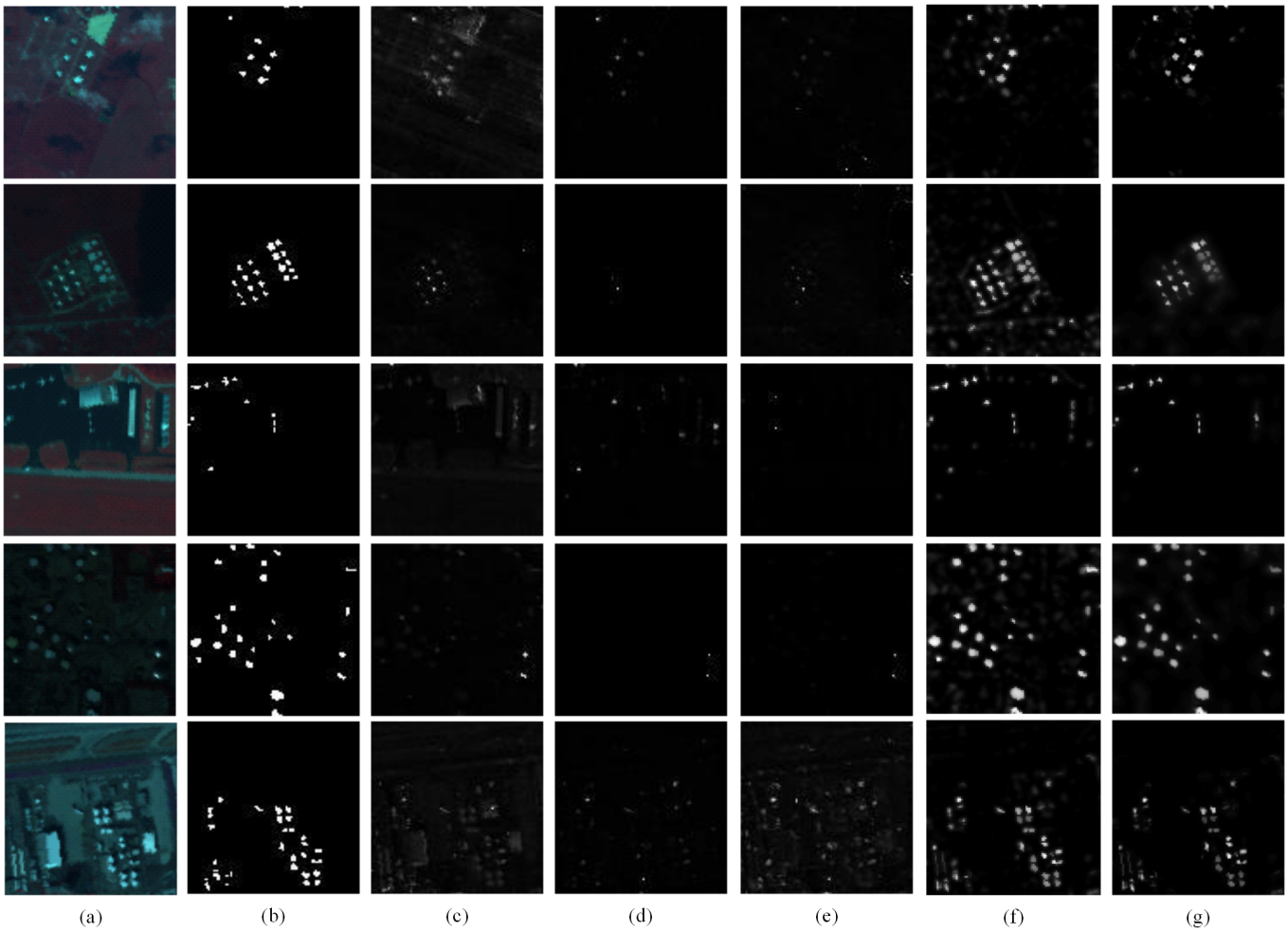


Fig. 7. Detection maps of different methods. (a) Color composites of ABU-urban scene, (b) reference map, (c) RXD, (d) LRXD, (e) CRD, (f) STGF, and (g) proposed.

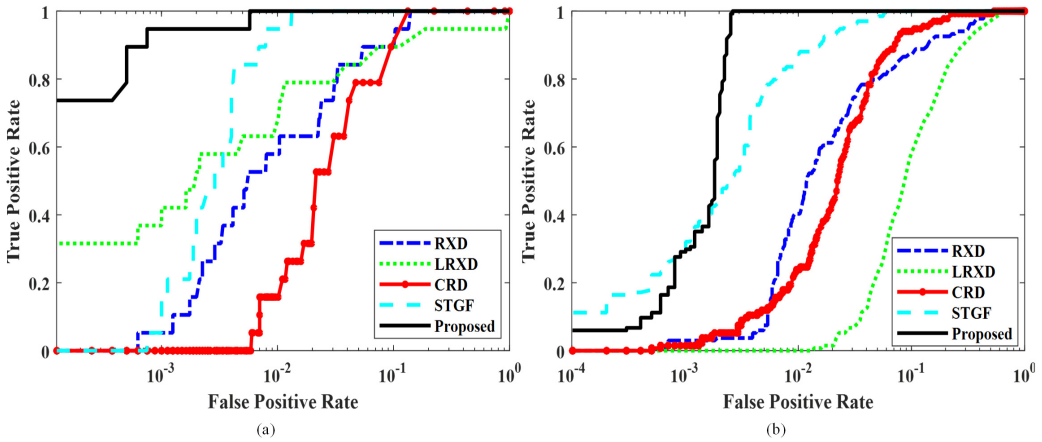


Fig. 8. ROC curves of (P_D, P_F) obtained by the compared methods on the (a) HYDICE and (b) San Diego data sets.

Furthermore, Fig. 8 illustrates the ROC curves of (P_D, P_F) obtained by different detectors using two typical and widely used data sets, i.e., HYDICE and San Diego. These two data sets are representative of anomalies at subpixel level and anomalies with full-size pixels, respectively. From the results, it can be clearly seen that the proposed detection method is always superior to the comparative methods. This result confirms that our method has promising results for these scenes, which also illustrates that the proposed detector has better generalization ability. The ROC curve of (P_F, t) representing the false alarm rate is illustrated in Fig. 9. It can be observed that the proposed method achieves smaller

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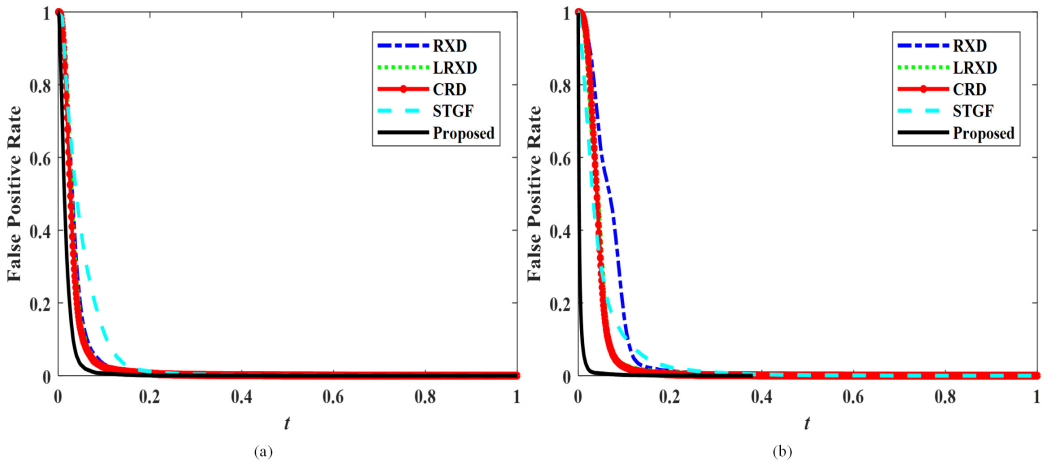


Fig. 9. ROC curves of (P_F, t) obtained by the compared methods on the (a) HYDICE and (b) San Diego data sets.

TABLE II
AVERAGE AUC VALUES OF (P_D, P_F) FOR THE COMPARED METHODS ON ALL DATA SETS

Image	RXD	LRXD	CRD	STGF	Proposed
ABU	0.94503	0.95178	0.93729	0.99612	**0.99744**
HYDICE	0.97637	0.93407	0.96206	0.99822	**0.99861**
San Diego	0.92844	0.78095	0.84199	0.98464	**0.99462**
EI Segundo	0.98153	0.94443	0.89549	0.98635	**0.99118**
Grand Isle	0.97401	0.85505	0.99013	0.98648	**0.99739**
Average	0.95564	0.80358	0.91149	0.98941	**0.99564**

TABLE III
AVERAGE AUC VALUES OF (P_F, t) FOR THE COMPARED METHODS ON ALL DATA SETS

Image	RXD	LRXD	CRD	STGF	Proposed
ABU	0.02536	0.02077	0.02194	0.05038	**0.00786**
HYDICE	0.03898	0.03498	0.03273	0.05335	**0.01859**
San Diego	0.05521	0.03810	0.08229	0.05106	**0.00617**
EI Segundo	0.00268	**0.00151**	0.00307	0.07500	0.01393
Grand Isle	0.01192	**0.00151**	0.00931	0.02134	0.00547
Average	0.03156	0.02250	0.03860	0.05036	**0.00969**

TABLE IV
AVERAGE AUC VALUES OF (P_D, P_F) FOR THE COMPARED BAND SELECTION METHODS ON ALL DATA SETS

Image	E-FDPC	OCF	DSEBS	SBSS	S-M	Proposed
ABU	0.95632	0.96296	0.95437	0.95608	0.94462	**0.99744**
HYDICE	0.98390	0.98434	0.98603	0.96890	0.99023	**0.99861**
San Diego	0.96371	0.97097	0.92777	0.96528	0.97727	**0.99462**
EI Segundo	0.94925	0.95481	0.96144	0.94275	0.91178	**0.99118**
Grand Isle	0.99289	0.99655	0.99619	0.99660	0.97699	**0.99739**
Average	0.96829	0.97343	0.95893	0.96581	0.96303	**0.99564**

TABLE V
AVERAGE AUC VALUES OF (P_F, t) FOR THE COMPARED BAND SELECTION METHODS ON ALL DATA SETS

Image	E-FDPC	OCF	DSEBS	SBSS	S-M	Proposed
ABU	0.01959	0.01781	0.02133	0.01751	0.02799	**0.00786**
HYDICE	0.02636	0.02263	0.02479	0.02324	0.03570	**0.01859**
San Diego	0.02646	0.02499	0.02232	0.02398	0.02699	**0.00617**
EI Segundo	0.00161	0.00164	0.00162	0.00162	**0.00157**	0.01393
Grand Isle	0.00682	**0.00436**	0.00609	**0.00436**	0.01111	0.00547
Average	0.01789	0.01607	0.01641	0.01578	0.02173	**0.00969**

AUC values of (P_F, t), which demonstrates that the proposed method can suppress background better.

C. Effect of Band Selection

To demonstrate the effectiveness of the proposed band selection approach, we replace it with either frequently cited in the literature or recent proposed methods. Five methods, i.e., enhanced density-peak-based clustering (E-FDPC) [43], optimal clustering framework (OCF) [44], dominant-set-extraction-based selector (DSEBS) [45], saliency bands and scale selection (SBSS) [46], and split-and-merge approach (S-M) [47], are used for comparison. For a fair comparison, the number of bands selected by different methods is the same. The AUC scores of the ROC curve of (P_D, P_F) and (P_F, t) are, respectively, presented in Tables IV and V. Our proposed band selection method achieves the best AUC scores of the ROC curve of (P_D, P_F) and (P_F, t) on average. It is concluded that the proposed band selection method shows competitive performances for hyperspectral anomaly detection in terms of detection accuracy and background suppression.

V. CONCLUSION

In this article, a novel approach to band selection for spectral–spatial anomaly detection is presented. With respect

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to hyperspectral anomaly detection, we first select a representative band subset based on the proposed physical constraint VAE in an unsupervised way. Intuitively, the selected band subset contains more discriminative features between anomalies and background than the discarded bands. Furthermore, the spatial detection result and spectral detection result are adaptively combined to decrease the miss detection rate. Finally, a simple iterative suppression is introduced to further reduce the false alarm rate. Compared to the existing methods, either the average detection performance or the false alarm rate produced by our proposed method is the best one. Besides, the proposed band selection method has the ability to select more informative yet more distinctive bands for hyperspectral anomaly detection in comparison to the state of the art, and it may be applied in the reflectance domain. In the future, how to implement the proposed method in real time and reduce parameters that need to be empirically assigned will be the focus.

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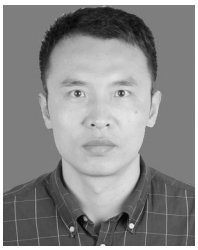
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